

9. When $y^2 + my + 14$ is divided by $(y - 3)$, the quotient is $f(y)$ and the remainder is k .
When $y^2 + my + 14$ is divided by $(y - 7)$, the quotient is $g(y)$ and the remainder is w .
If $w = 2k + 1$, find the value of m .
10. All lengths of the sides of a triangle are integers. The cosine of the smallest angle of the triangle is $\frac{19}{21}$. The length of one of the adjacent sides of the triangle for that angle is 6.
Find the smallest perimeter for such a triangle.
11. From the set $\{A, B, C, D, E, F\}$, two letters are selected at random without replacement.
Find the probability that A or B or both A and B were selected. Express your answer as a common fraction reduced to lowest terms.
12. If $\ln$ is the symbol for natural logarithm, find the value of y such that
 $\ln(5 + 4y) - \ln(3 + y) = \ln(3)$.
13. In Triangle ABC , $AB = 64$, $BC = 64$, and $AC = 48$. A , B , and C are the centers of three mutually externally tangent circles whose points of tangency are D , E , and F .
Rounded to the nearest **whole number**, find the perimeter of Triangle DEF .
14. Last year UIC traced the job placement of their high school mathematics teacher candidates and found all took jobs teaching. The arithmetic mean salary of the candidates who took jobs in Chicago was \$48,600. The arithmetic mean of the rest of the candidates, who took jobs outside Chicago, was \$43,800. The overall arithmetic mean for all these candidates was \$45,000. Find the fraction of the candidates who took jobs outside Chicago. Express your answer as a common fraction reduced to lowest terms.